ARX-7 Collaborative Robot Arm

User Manual - Version 2.3

Vantor Robotics

About this manual

This manual describes the safe installation, operation, programming and maintenance of the ARX-7 collaborative robot arm manufactured by Vantor Robotics. It is intended for integrators, operators and maintenance technicians. Read the safety chapter completely before powering on the robot for the first time.

Vantor Robotics was founded in 2014 and develops lightweight collaborative robots for small and medium-sized manufacturers. The ARX-7 was first released in 2021 and is the company's best-selling model, mainly used for machine tending, packaging, screwdriving and quality inspection.

Note: Vantor Robotics and the ARX-7 are fictional. This document was created for testing a document question-answering system.

1. Safety

The ARX-7 is designed to work alongside people without a safety fence, but only after a risk assessment of the complete application has been carried out by the integrator. The robot alone is not a complete machine; the gripper, the workpiece and the surrounding equipment can all introduce hazards.

1.1 Built-in safety functions

- **Collision detection:** the robot stops when the measured joint torque differs from the expected value by more than the configured limit. The default collision sensitivity is level 3 on a scale from 1 (least sensitive) to 5 (most sensitive).
- **Speed limiting:** in collaborative mode the tool speed is limited to 250 mm/s. In reduced mode, triggered by a safety scanner, the limit is 100 mm/s.
- **Emergency stop:** the red emergency stop button on the teach pendant triggers a Category 1 stop - the robot decelerates in a controlled way and then power to the motors is removed.
- **Protective stop:** an external safety input that pauses the program. Motion can resume after the input is cleared and the operator presses Resume.

1.2 Operator rules

- Never place your head or face inside the robot's working area while a program is running.
- Do not use the robot to handle sharp objects in collaborative mode without additional protection.
- Test the emergency stop button at the start of every shift.
- Only trained personnel may change safety settings. Safety settings are protected by a separate safety password, which is different from the normal administrator password.

2. Technical Specifications

The table below lists the main technical data of the ARX-7. All values are measured at an ambient temperature of 25 °C.

Parameter	Value
Degrees of freedom	6 rotating joints
Maximum payload	7 kg
Reach	850 mm
Pose repeatability	+/- 0.03 mm
Robot arm weight	24 kg
Maximum tool speed	1.5 m/s (non-collaborative mode)
Joint ranges	+/- 360 deg for all joints except joint 3 (+/- 165 deg)
Protection rating	IP54 (arm), IP20 (control box)
Power supply	100-240 V AC, 50/60 Hz
Typical power consumption	180 W during a typical pick-and-place cycle
Operating temperature	0 - 45 deg C
Noise level	below 65 dB(A)
Communication	Ethernet, Modbus TCP, PROFINET, 16 digital I/O
Mounting	Floor, wall or ceiling, any angle

The maximum payload of 7 kg includes the weight of the gripper. For example, if the gripper weighs 1.2 kg, the robot can carry a workpiece of up to 5.8 kg. At full reach the recommended payload is reduced to 5 kg to maintain accuracy.

3. Installation and Setup

3.1 Mounting surface

The mounting surface must withstand at least 10 times the torque of joint 1 at full speed, which corresponds to approximately 1,500 Nm. A steel plate at least 15 mm thick or a concrete floor is recommended. The robot base is fixed with four M8 bolts tightened to 20 Nm.

3.2 First start-up

- Connect the robot arm cable to the control box before connecting mains power.
- Switch on the control box. The status LED blinks blue while the system boots, which takes about 40 seconds.
- When the LED turns solid green, the robot is ready and the teach pendant shows the initialization screen.
- Release the joint brakes by pressing **Initialize**. You will hear a short click from each joint.
- Set the payload and the tool center point (TCP) in Settings > Tool before running any program.

3.3 Default credentials

The default administrator password is **vantor123**. You will be asked to change it during first login. The default IP address of the control box is **192.168.1.50**.

3.4 Environment

Do not install the robot in environments with explosive atmospheres, strong magnetic fields or direct exposure to cutting fluid spray. Relative humidity must stay between 10 % and 90 % without condensation.

4. Operating Modes and Programming

4.1 Operating modes

The ARX-7 has three operating modes. **Manual mode** is used for teaching positions and limits speed to 250 mm/s. **Automatic mode** runs programs at full programmed speed and requires all safety inputs to be active. **Remote mode** lets an external PLC or computer start and stop programs over Modbus TCP or PROFINET.

4.2 Teaching by demonstration

Press and hold the **Freedrive** button on the back of the teach pendant to move the arm by hand. Release the button to lock the position, then tap **Add Waypoint** to store it. Up to 2,000 waypoints can be stored in a single program.

4.3 VantorScript

For advanced applications the robot can be programmed in VantorScript, a Python-like scripting language. Scripts are written in the Script editor on the teach pendant or uploaded over Ethernet. The most important commands are:

- **move_joint(q, speed)** - moves to a joint position; fastest motion, path not linear.
- **move_linear(pose, speed)** - moves the tool in a straight line, used for precise insertion.
- **set_output(n, value)** and **get_input(n)** - control digital I/O, for example a pneumatic gripper.
- **wait(seconds)** - pauses the program.

Programs are saved with the extension **.vsp**. The control box stores up to 500 programs internally; more can be kept on a USB drive.

5. Troubleshooting and Error Codes

When an error occurs, the status LED turns red and the teach pendant shows an error code. The most common codes are listed below.

Code	Meaning	What to do
E101	Collision detected	Remove the obstacle, check the payload setting, then press Resume.
E204	Joint position limit exceeded	Move the joint back into range in Manual mode using Freedrive.
E310	Joint overheating	Stop the robot for at least 15 minutes and reduce the cycle speed or payload.
E415	Communication lost with joint	Check the arm cable connector at the control box; restart the control box.
E502	Emergency stop pressed	Release the emergency stop button by turning it clockwise, then press Reset.
E640	Payload mismatch	The measured load differs from the configured payload by more than 0.5 kg. Update Settings > Tool.
E777	Safety configuration checksum error	Contact Vantor Robotics support. Do not operate the robot.

If the same error appears three times within one hour, export the log file (Settings > System > Export Logs) to a USB drive and send it to support together with the robot serial number.

6. Maintenance, Warranty and Support

6.1 Maintenance schedule

Interval	Task
Every shift	Test the emergency stop button; visually check cables and gripper.
Every month	Clean the arm with a dry cloth; check that base bolts are tight (20 Nm).
Every 6 months	Check joint brakes using the Brake Test function in Settings > Maintenance.
Every 20,000 operating hours	Replace joint gearboxes - service performed by Vantor Robotics only.
Every 2 years	Replace the control box backup battery (type CR2032).

6.2 Warranty

The ARX-7 comes with a warranty of 24 months from the date of delivery or 8,000 operating hours, whichever comes first. The warranty does not cover damage caused by exceeding the maximum payload, by unauthorized changes to safety settings, or by grippers and tools from other manufacturers. Warranty can be extended to 48 months with the Vantor CarePlus service contract.

6.3 Support

Technical support is available Monday to Friday from 08:00 to 17:00 CET by email at support@vantor-robotics.example or by phone at +43 316 555 0170. Please have the robot serial number ready; it is printed on the label at the bottom of the robot base and starts with the letters VX.